

Workflow	Time (h)	Medal Rate (%)			Overall Metrics (%)			Mean Percentile (%)
		Lite (22)	Medium (38)	High (15)	All (75)	Gold	Med+	
Single Agents with Official Harnesses (/goal)								
Claude Opus 5	6	80.3 ± 1.5	78.1 ± 2.3	46.7 ± 0.0	72.4 ± 1.6	50.7 ± 0.8	85.3 ± 1.3	84.8 ± 0.6
GPT-5.6-sol	6	80.3 ± 1.5	69.3 ± 2.3	46.7 ± 6.7	68.0 ± 1.3	47.6 ± 2.5	81.3 ± 0.8	81.3 ± 1.3
Open-Source Agent Baselines								
ScienceFlow	24	80.3 ± 1.5	74.6 ± 0.9	44.4 ± 2.2	70.2 ± 1.2	—	—	—
DeepSeek V4 Flash-preview								
MLEvolve	12	80.3 ± 1.5	64.0 ± 0.9	46.7 ± 0.0	65.3 ± 0.8	34.7 ± 0.0	76.0 ± 2.3	—
Gemini-3.1-Pro-preview								
HMA (Ours)								
Opus 5 ↔ GPT-5.6-sol								
Avg. of Single	6	80.3 ± 1.1	73.7 ± 1.6	46.7 ± 3.3	70.2 ± 1.0	49.1 ± 1.3	83.3 ± 0.8	83.0 ± 0.7
HMA (ours)	6	84.8 _{+4.5}	80.7 _{+7.0}	62.2 _{+15.6}	78.2 _{+8.0}	53.8 _{+4.7}	90.7 _{+7.3}	89.0 _{+5.9}
GPT-5.6-sol ↔ DeepSeek V4.1 Flash								
Avg. of Single	6	78.8 ± 1.5	65.4 ± 3.4	42.2 ± 3.5	64.7 ± 2.1	42.7 ± 1.3	77.8 ± 1.2	78.0 ± 1.3
HMA (ours)	6	81.8 _{+3.0}	76.3 _{+10.9}	46.7 _{+4.5}	72.0 _{+7.3}	50.7 _{+8.0}	85.3 _{+7.6}	84.4 _{+6.4}
GPT-5.6-sol ↔ Kimi K3								
Avg. of Single	6	78.8 ± 1.5	57.5 ± 1.6	44.4 ± 3.5	61.1 ± 1.0	41.1 ± 1.4	77.3 ± 1.2	77.6 ± 1.0
HMA (ours)	6	81.8 _{+3.0}	71.1 _{+13.6}	53.3 _{+8.9}	70.7 _{+9.6}	50.7 _{+9.6}	85.3 _{+8.0}	83.9 _{+6.4}